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The Evolution of Artificial Intelligence

Seventy-five years of ambition, collapse, and breakthrough — from Alan Turing's question "Can machines think?" to the generative AI race reshaping every industry.

Foundations | AI Winters | Expert Systems Boom | Quiet Progress | Deep Learning Era | Generative AI Era

1950

Turing asks: "Can machines think?"

Alan Turing publishes *Computing Machinery and Intelligence*, proposing the "imitation game" — the Turing Test — as a practical measure of machine intelligence. The wartime code-breaker becomes the field's founding thinker.

Key players: Alan Turing

1956

The Dartmouth Workshop — AI gets its name

John McCarthy coins the term "artificial intelligence" at a summer workshop at Dartmouth College. The founders predict machines will match human intelligence within a generation — optimism that will define, and haunt, the field.

Key players: **John McCarthy, Marvin Minsky, Claude Shannon, Nathaniel Rochester**

1957

The Perceptron — the first neural network

Frank Rosenblatt builds the Perceptron, a machine that learns from data by adjusting weighted connections — the ancestor of every modern neural network. The New York Times reports it may one day "walk, talk, see, and write."

Key players: **Frank Rosenblatt**

1966

ELIZA — the first chatbot

Joseph Weizenbaum's ELIZA mimics a psychotherapist using simple pattern matching. Users pour their hearts out to it — an early warning of how easily humans attribute understanding to machines.

Key players: Joseph Weizenbaum (MIT)

1969

"Perceptrons" exposes the limits

Minsky and Papert prove mathematically that single-layer perceptrons cannot solve simple problems like XOR. Neural network research funding dries up almost overnight — symbolic, logic-based AI dominates instead.

Key players: Marvin Minsky, Seymour Papert

❄ The First AI Winter (1974–1980)

Grand promises collide with tiny computers. The UK's Lighthill Report (1973) concludes AI has failed to deliver; DARPA slashes funding in the US. **Cause:** hype outran hardware — there simply wasn't enough computing power. **Impact:** research funding and talent drain from the field for half a decade.

1980

Expert systems bring AI to business

Rule-based "expert systems" encode human specialist knowledge. DEC's XCON system saves the company an estimated $40M a year configuring computer orders, and corporations worldwide rush to build their own.

Key players: **Edward Feigenbaum, Digital Equipment Corporation**

1982

Japan's Fifth Generation Project

Japan commits $850M to build intelligent computers, triggering competitive AI funding programs in the US and UK. Symbolic AI reaches its commercial peak.

Key players: **Japan's MITI**

1986

Backpropagation revives neural networks

Rumelhart, Hinton, and Williams popularize backpropagation, showing how to train *multi-layer* networks — solving the very limitation Minsky exposed in 1969. Connectionism quietly returns from exile.

Key players: **David Rumelhart, Geoffrey Hinton, Ronald Williams**

❄ The Second AI Winter (1987–1993)

The specialized LISP machine market collapses as ordinary desktop computers overtake it, and expert systems prove brittle and expensive to maintain. **Cause:** rigid rule-based systems couldn't adapt or scale. **Impact:** "AI" becomes a dirty word in funding proposals; researchers rebrand their work as "machine learning" and "informatics."

1997

Deep Blue defeats the world chess champion

IBM's Deep Blue beats Garry Kasparov — the first machine victory over a reigning world champion under tournament conditions. Brute-force search plus specialized hardware puts AI back on the world's front pages.

Key players: IBM, Garry Kasparov

2006

"Deep learning" is born — and the cloud arrives

Geoffrey Hinton shows deep belief networks can be trained layer by layer, rebranding neural nets as "deep learning." The same year, Amazon launches AWS — the cloud computing infrastructure that will let anyone rent massive compute on demand.

Key players: Geoffrey Hinton, Amazon Web Services

2009

ImageNet — big data for vision

Fei-Fei Li's team publishes ImageNet, a dataset of millions of labeled images. Its annual competition becomes the proving ground where deep learning will stage its coup.

Key players: Fei-Fei Li (Stanford)

2012

AlexNet — the moment hardware caught up

A deep neural network trained on Nvidia gaming GPUs crushes the ImageNet competition, nearly halving the error rate. Theory from the 1980s finally meets sufficient hardware — and the deep learning gold rush begins.

Key players: Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton, Nvidia

2016

AlphaGo beats Lee Sedol

DeepMind's AlphaGo defeats the legendary Go champion 4–1, mastering a game with more board positions than atoms in the universe. Its "Move 37" — a move no human would play — shows machines can be creative, not just fast.

Key players: **DeepMind (Google), Demis Hassabis, Lee Sedol**

2017

"Attention Is All You Need" — the Transformer

Google researchers publish the Transformer architecture, which processes language in parallel rather than word by word. It becomes the foundation of virtually every modern AI model — the "T" in GPT.

Key players: **Vaswani et al. (Google Brain)**

2020

GPT-3 — scale becomes a strategy

OpenAI's 175-billion-parameter GPT-3 writes essays, code, and poetry from a plain-text prompt. Training compute for frontier models is now doubling roughly every six months — far faster than Moore's Law.

Key players: OpenAI, Sam Altman

2022

ChatGPT — AI goes mainstream

ChatGPT reaches 100 million users in two months — the fastest-adopted consumer application in history at the time. Generative AI leaves the lab and lands on everyone's desk, desk drawer, and dinner-table conversation.

Key players: OpenAI

2023–25

The generative AI race

GPT-4, Google's Gemini, Anthropic's Claude, and open models like Llama compete on multimodal understanding and step-by-step reasoning. Nvidia becomes one of the world's most valuable companies — hardware and algorithms, as always, advancing together.

Key players: **OpenAI, Google, Anthropic, Meta, Nvidia**

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An original visual timeline created by **Dominic Ndondo** for AIML-500 (Machine Learning Fundamentals), Indiana Wesleyan University — July 2026. Research was assisted by AI tools (Anthropic's Claude), per the assignment's guidance on disclosing AI assistance; all events were cross-checked against the sources above. Designed and hand-coded in HTML & CSS. Part of my **professional portfolio**.